

Report

Dataset: iSDA Soil Properties for Sub-Saharan Africa

1: Data Sourcing

Dataset Source: iSDA (Innovative Solutions for Decision Agriculture) Soil Properties dataset, publicly available via [Kaggle](#).

Dataset Description: 49,225 georeferenced soil samples across Sub-Saharan Africa. Each record contains 24 variables spanning location (longitude, latitude), sampling metadata (start_date, end_date, source, horizon depth), and 19 soil chemical properties measured via Mehlich3 extraction and combustion methods.

Selection Rationale: This dataset includes the full NPK suite (nitrogen_total, potassium_extractable, phosphorus_extractable), pH, and micronutrients. The 49K sample size provides statistical power for robust recommendation logic. Unlike smaller crop-yield datasets, this gives ground-truth soil chemistry rather than aggregated regional averages.

Expected Impact on Recommendations: Enables location-specific fertilizer and crop recommendations. The horizon depth data (topsoil vs subsoil) allows differentiated advice for shallow-rooted vs deep-rooted crops. The aluminium_extractable and pH pairing flags toxicity risk zones where lime amendment is critical before planting.

2: Data Cleaning

Issues Found:

- 42 duplicate rows identified across all 24 columns.
- Six columns with >95% structural missingness: boron_extractable (4.3% coverage), electrical_conductivity (4.3%), phosphorus_extractable (4.3%), sodium_extractable (4.4%), sulphur_extractable (4.3%), zinc_extractable (4.3%). These are wet-chemistry variables with sparse lab analysis.
- Scattered nulls in aluminium_extractable (2 missing), ph (3 missing), carbon_total (2,157 missing), copper_extractable (46 missing), iron_extractable (46 missing), manganese_extractable (48 missing), nitrogen_total (46 missing).
- Extreme outliers in calcium_extractable (max 32,376 mg/kg vs median 997), carbon_organic (max 310 vs median 9.93), and potassium_extractable (max 3,830 vs median 177).

Cleaning Steps Taken:

- Dropped 42 duplicate rows. Dataset reduced from 49,225 to 49,183.

- Dropped six structurally missing columns (boron_extractable, electrical_conductivity, phosphorus_extractable, sodium_extractable, sulphur_extractable, zinc_extractable). Retained 15 columns.
- Imputed carbon_total missing values with median (10.7) to preserve 2,157 records. Dropped remaining 46 scattered nulls across aluminium, ph, copper, iron, manganese, nitrogen. Final clean dataset: 47,023 rows x 15 columns.
- Engineered horizon categorical: coded horizon_upper == 0 as topsoil, else subsoil. Dropped horizon_upper and horizon_lower. Final schema: 14 columns.
- Capped outliers via Interquartile Range (IQR) method on carbon_organic, calcium_extractable, aluminium_extractable, potassium_extractable, magnesium_extractable. Upper bounds set at $Q3 + 1.5 \times IQR$.

Cleaned Dataset: Exported as **cleaned_dataset.csv** (47,023 rows x 14 columns). Zero nulls. All numeric columns within biologically plausible ranges.

3: Exploratory Data Analysis

Five Visualizations and Insights:

Visualization 1: Distribution of Key Soil Nutrients (histogram grid)

pH shows near-normal distribution centered at 6.3. Nitrogen_total, carbon_organic, potassium_extractable, and calcium_extractable are all right-skewed with long tails. This means mean values exceed medians; recommendation logic must use median thresholds rather than means to avoid overestimating soil fertility.

Visualization 2: Topsoil vs Subsoil Nutrient Comparison (boxplots)

Topsoil carries higher nitrogen_total and carbon_organic than subsoil. pH is nearly identical across depths. This validates surface fertilization strategies for shallow-rooted crops (vegetables, groundnuts) and suggests deep-rooted crops (cassava, sorghum) may need subsoil amendment.

Visualization 3: Soil pH Distribution Across Sub-Saharan Africa (geosscatter)

Acidic soils ($pH < 5.5$) cluster in the equatorial belt of East and West Africa. Alkaline soils ($pH > 7.5$) concentrate in the Horn of Africa and southern tip. East Africa shows the densest sampling and widest pH range

Visualization 4: Correlation Matrix of Soil Nutrients (heatmap)

Carbon_organic and nitrogen_total correlate at $r=0.88$. Calcium_extractable and magnesium_extractable correlate at $r=0.89$. Calcium and pH correlate at $r=0.78$. These collinearities mean the recommendation engine can reduce dimensionality: carbon_organic proxies for nitrogen status; calcium proxies for both magnesium and pH buffering.

Visualization 5: Aluminium Toxicity Risk by pH (scatter with threshold)

Below pH 5.5, aluminium_extractable spikes to >1,500 mg/kg. Above pH 6.0, aluminium drops below 600 mg/kg. The pH 5.5 threshold is a hard boundary for toxicity-sensitive crops (beans, maize). Flag any sample below this threshold for lime amendment before recommending sensitive crops.

4: Recommendation Feature Engineering

Feature 1: Soil Organic Matter (SOM)

Formula: $\text{carbon_organic} \times 1.724$ (Van Bemmelen factor).

Usefulness: SOM is the agronomic standard for soil health, not raw carbon.

Expected Impact: Enables direct comparison against FAO soil fertility classes (very low < 0.5%, low 0.5-1.5%, medium 1.5-3%, high > 3%). We can now tell a farmer their SOM class, not just a carbon number.

Practicality: Single arithmetic operation. No external data is needed. Instant computation.

Feature 2: Aluminium Toxicity Flag

Formula: $(\text{aluminium_extractable} > 1,000 \text{ mg/kg}) \text{ AND } (\text{pH} < 5.5) \rightarrow 1, \text{ else } 0.$

Usefulness: Binary risk indicator. Removes ambiguity from raw aluminium numbers.

Expected Impact: 4,055 of 47,023 samples (8.6%) flag positive. We can auto-block aluminium-sensitive crop recommendations (maize, beans) on these sites and push lime amendment or tolerant crops (cassava, certain sorghum varieties).

Practicality: Two threshold comparisons with no model training. Immediate deployment.

Feature 3: Is Topsoil Binary

Formula: $(\text{horizon} == \text{topsoil}) \rightarrow 1, \text{ else } 0.$

Usefulness: Enables depth-differentiated recommendations in the same field. Topsoil and subsoil have different nutrient profiles (Visualization 2).

Expected Impact: Weight fertilizer placement. Topsoil gets surface broadcast; subsoil gets deep placement or banding. This turns a single sample into a layered recommendation.

Practicality: Derived from existing horizon coding. Zero cost.

5: Recommendation Logic

Approach: Rule-based scoring system. No advanced modeling applied. Four soil parameters checked against crop-specific thresholds. Each match adds one point. Highest score wins. Ties broken by first match in dictionary order.

Crop Threshold Reference Table:

Crop	pH Range	N Min (%)	K Min (mg/kg)	Al Max (mg/kg)
Maize	5.5 - 7.5	0.8	80	500
Cassava	4.5 - 7.0	0.5	60	800
Sorghum	5.5 - 8.0	0.6	70	600
Groundnuts	5.0 - 7.0	0.4	50	700
Beans	6.0 - 7.5	0.7	80	400

Example Run: Input: pH=5.3, N=0.6%, K=95 mg/kg, Al=620 mg/kg.
Scores: Maize=1, Cassava=4, Sorghum=2, Groundnuts=4, Beans=1.
Recommended Crop: Cassava (tied with Groundnuts at 4 points; Cassava wins dictionary order).

Why This Works: Farmers need explainable advice. A black-box model fails here. The rule system gives transparent justification: Cassava scored 4 because pH 5.3 falls within its 4.5-7.0 range, N exceeds 0.5, K exceeds 60, and Al 620 is below its 800 tolerance. All points auditable.

Bonus: LLM-Assisted Insights

Prompt Used: "Given a soil dataset(cleaned_dataset attached) for Sub-Saharan Africa with columns for longitude, latitude, pH, nitrogen_total, carbon_organic, potassium_extractable, calcium_extractable, magnesium_extractable, aluminium_extractable, iron_extractable, manganese_extractable, and horizon depth, what are the top 3 agronomic insights that an agriculture startup that champions recommender systems for farmer should act on? Cite specific numerical thresholds from the data."

Insights Obtained:

- **Prioritize soil acidity management:** Approximately 25% of soils have pH < 5.86, indicating acidic conditions that can limit nutrient availability. Flag these fields for lime application and acid-tolerant crop recommendations.
- **Focus on rebuilding soil fertility:** The lowest 25% of soils have total nitrogen < 0.52 g/kg and organic carbon < 8.6 g/kg, suggesting poor soil health.

- Recommend integrated nutrient management (organic amendments, legumes, and balanced nitrogen fertilization) rather than nitrogen fertilizer alone.
- **Target potassium applications precisely:** Extractable potassium ranges widely, with the bottom 25% of soils below 105.8 mg/kg and a median of 184.8 mg/kg. Recommend potassium fertilizer only for low-K soils, avoiding unnecessary input costs for farmers with adequate potassium levels.

Usefulness Assessment: The agronomic insights reinforced the relevance of engineering features by highlighting soil acidity ($\text{pH} < 5.86$), low soil fertility (nitrogen < 0.52 g/kg and organic carbon < 8.6 g/kg), and potassium deficiency (extractable K < 105.8 mg/kg) as the primary decision thresholds for farmer recommendations. These findings support the feature engineering process by validating that the selected variables capture meaningful agronomic conditions requiring different management actions. The LLM primarily served as a validation tool rather than a discovery tool that confirmed that the engineered features and threshold-based recommendations were consistent with established agronomic principles without identifying fundamentally new patterns in the data.

Conclusion: The iSDA dataset provides a robust foundation for a recommendation engine. After cleaning, 47,023 high-quality samples remain. The EDA reveals clear agronomic boundaries (pH 5.5 toxicity threshold, topsoil-subsoil differentiation, carbon-nitrogen collinearity). Three engineered features (SOM, toxicity flag, topsoil binary) feed directly into a transparent rule-based recommender. The LLM bonus confirmed that human-engineered features matched domain expertise